DATA607 Lab 6

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Fall 2025

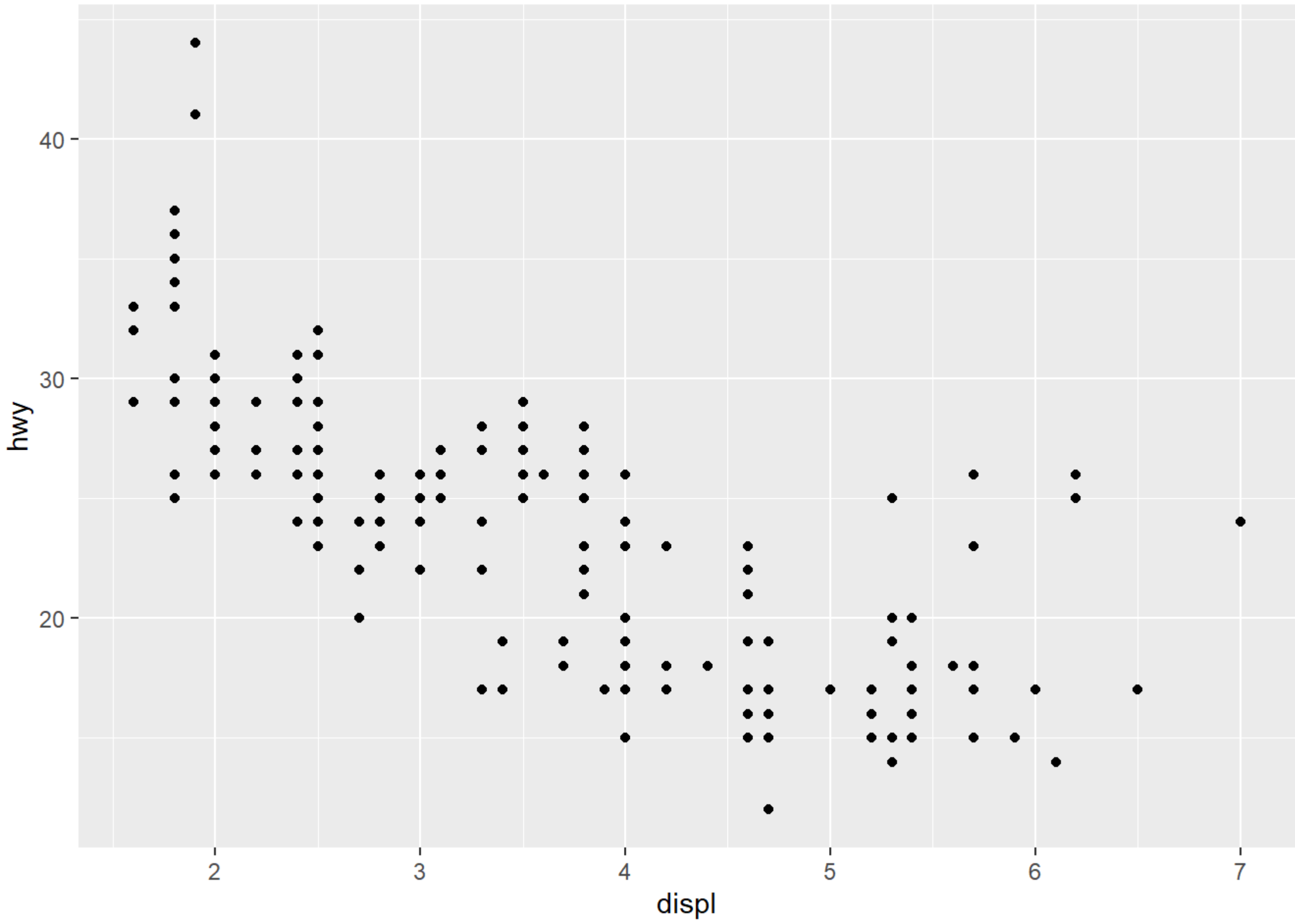
```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.5
## ✔ forcats   1.0.0     ✔ stringr   1.5.2
## ✔ ggplot2   4.0.0     ✔ tibble    3.3.0
## ✔ lubridate 1.9.4     ✔ tidyr     1.3.1
## ✔ purrr     1.1.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
come errors
```

```
#library(datasets)
str(mpg)
```

```
## tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
##  $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
##  $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
##  $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
##  $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
##  $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
##  $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
##  $ drv         : chr [1:234] "f" "f" "f" "f" ...
##  $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
##  $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
##  $ fl          : chr [1:234] "p" "p" "p" "p" ...
##  $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

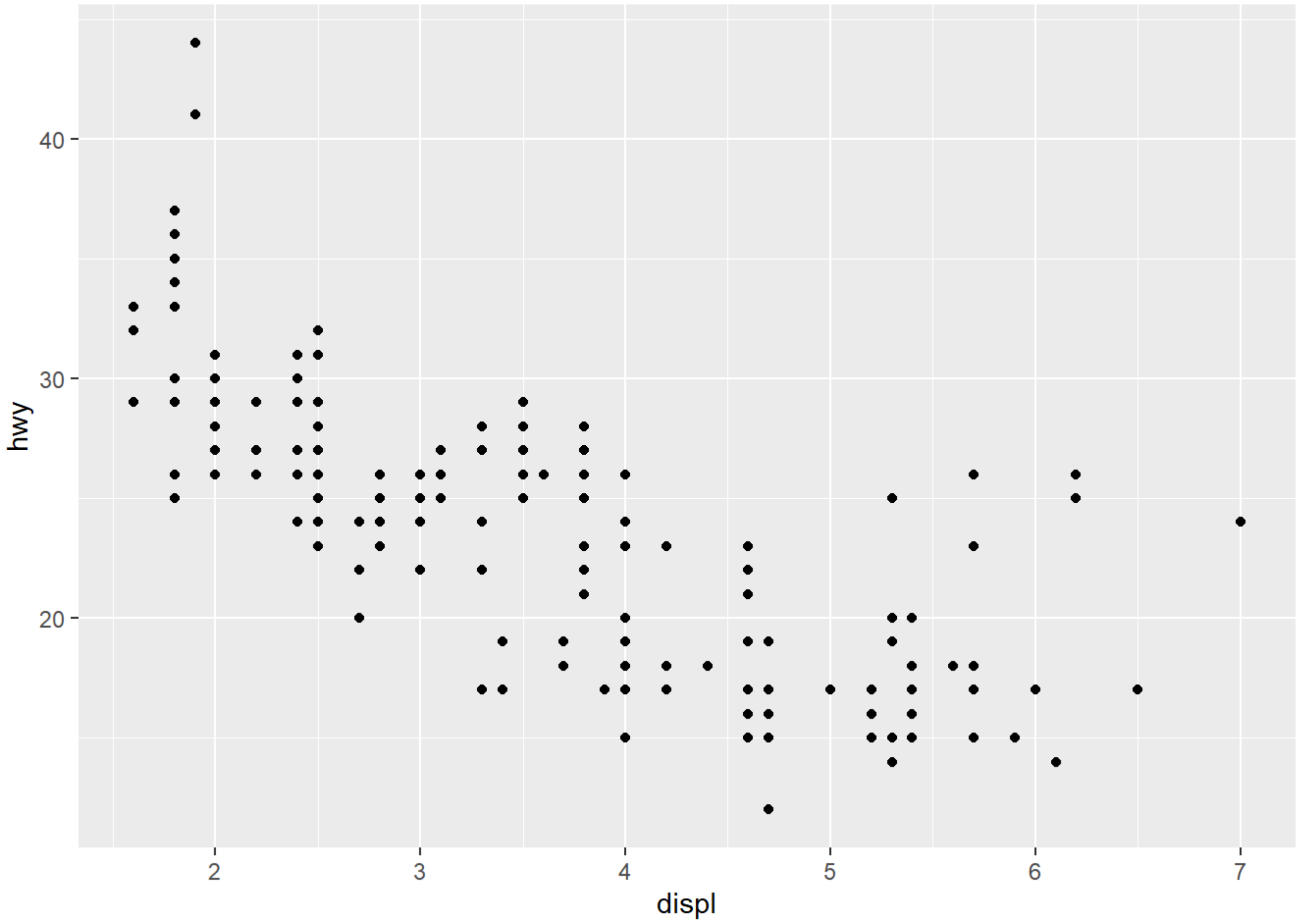
```
#Multiple ways to get the same thing done.
#The long way
ggplot() +
layer(
data = mpg, mapping = aes(x = displ, y = hwy),
geom = "point", stat = "identity", position = "identity"
) +
scale_y_continuous() +
scale_x_continuous() +
coord_cartesian()
```



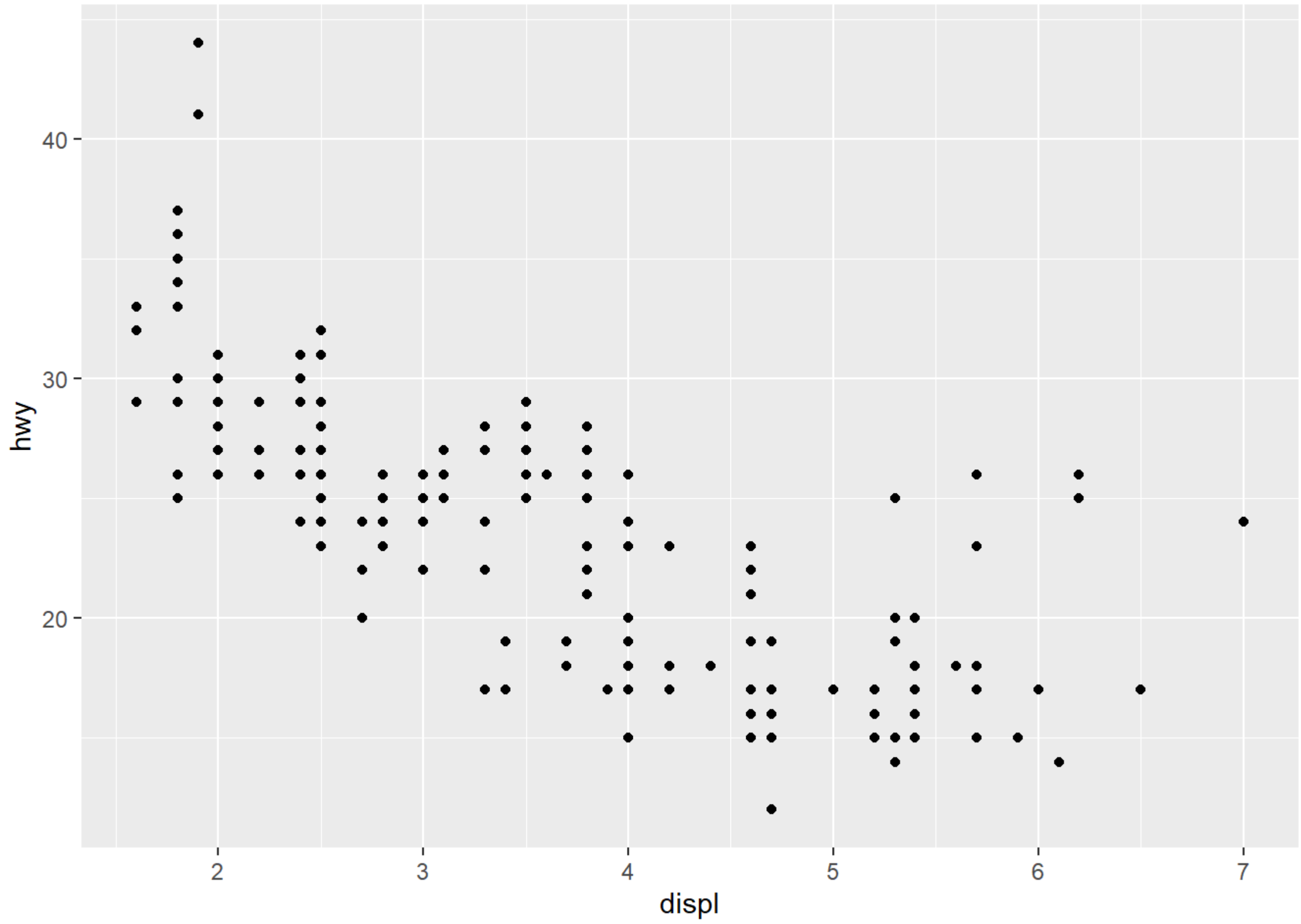
What are the drawbacks of the code above, specially in a situation where we are doing EDA ?

Ans - The drawback of the above code is that it is very long and some lines are redundant. When we are doing EDA in a real life scenario we are expected to plot a large number of plots and we would want a much shorter way to write the above code.

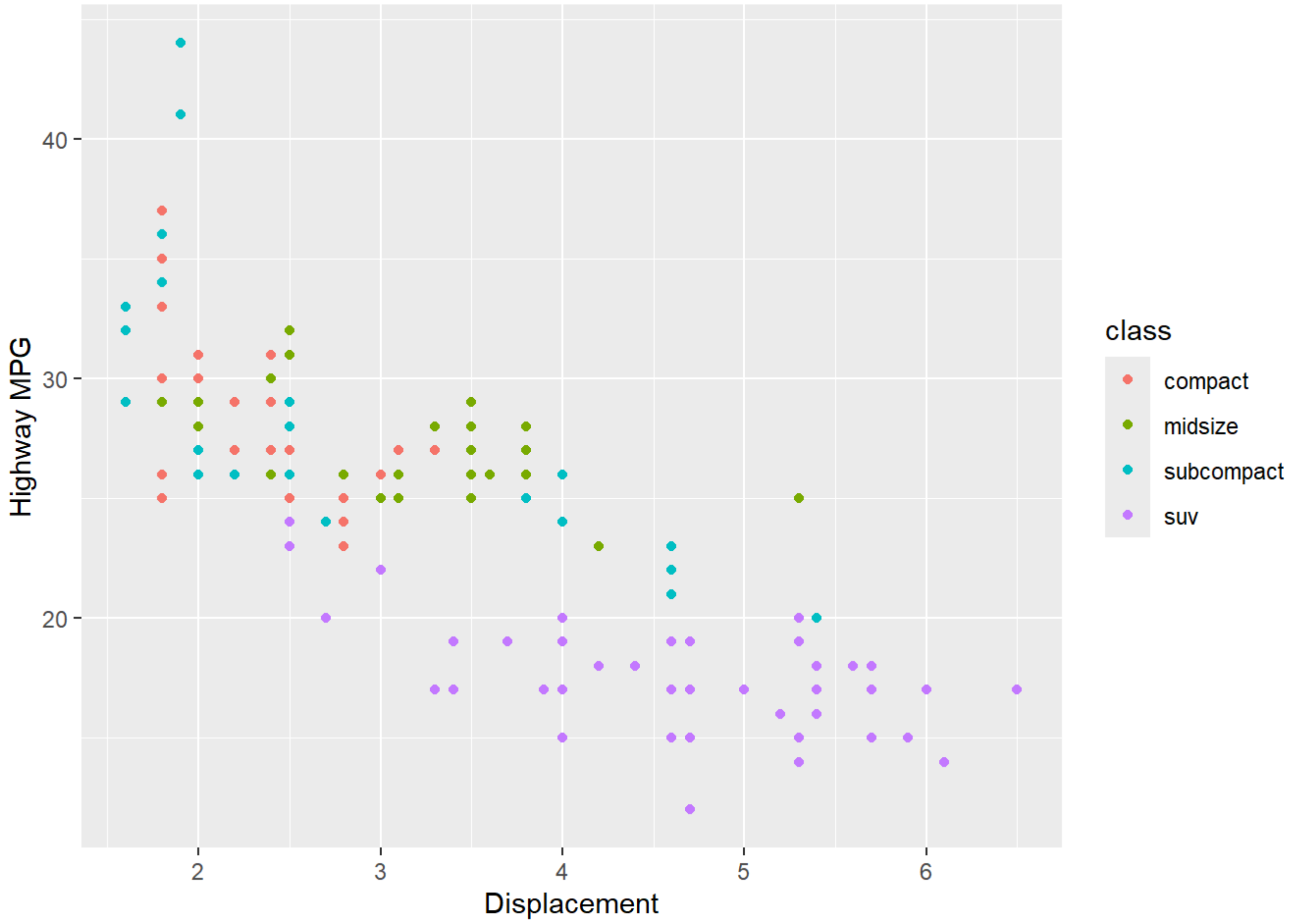
```
# Shorthand 1
g <- ggplot(data = mpg, aes(x=displ, y=hwy))
g + geom_point()
```




```
#Shorthand 2
g <- ggplot(data = mpg)
g + geom_point(aes(x=displ, y=hwy))
```



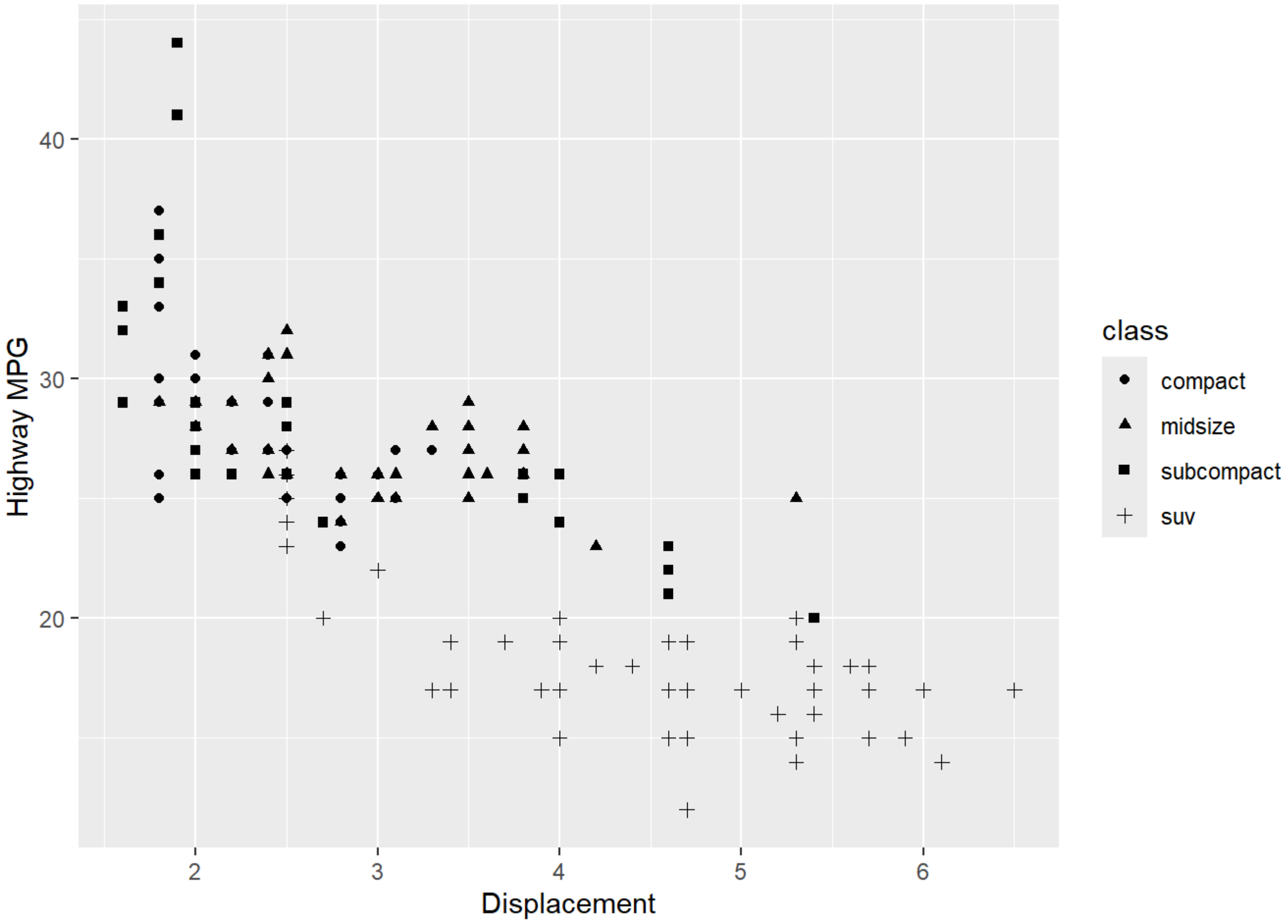

```
# Let's tweak things around!
# What does class %in% c(...) do?
# What does labs(....) do?
g <- ggplot(data = subset(mpg,
class %in% c("compact",
"midsize", "subcompact",
"suv")))
g + geom_point(aes(x=displ,
y=hwy, color = class)) + labs(x="Displacement", y="Highway MPG")
```



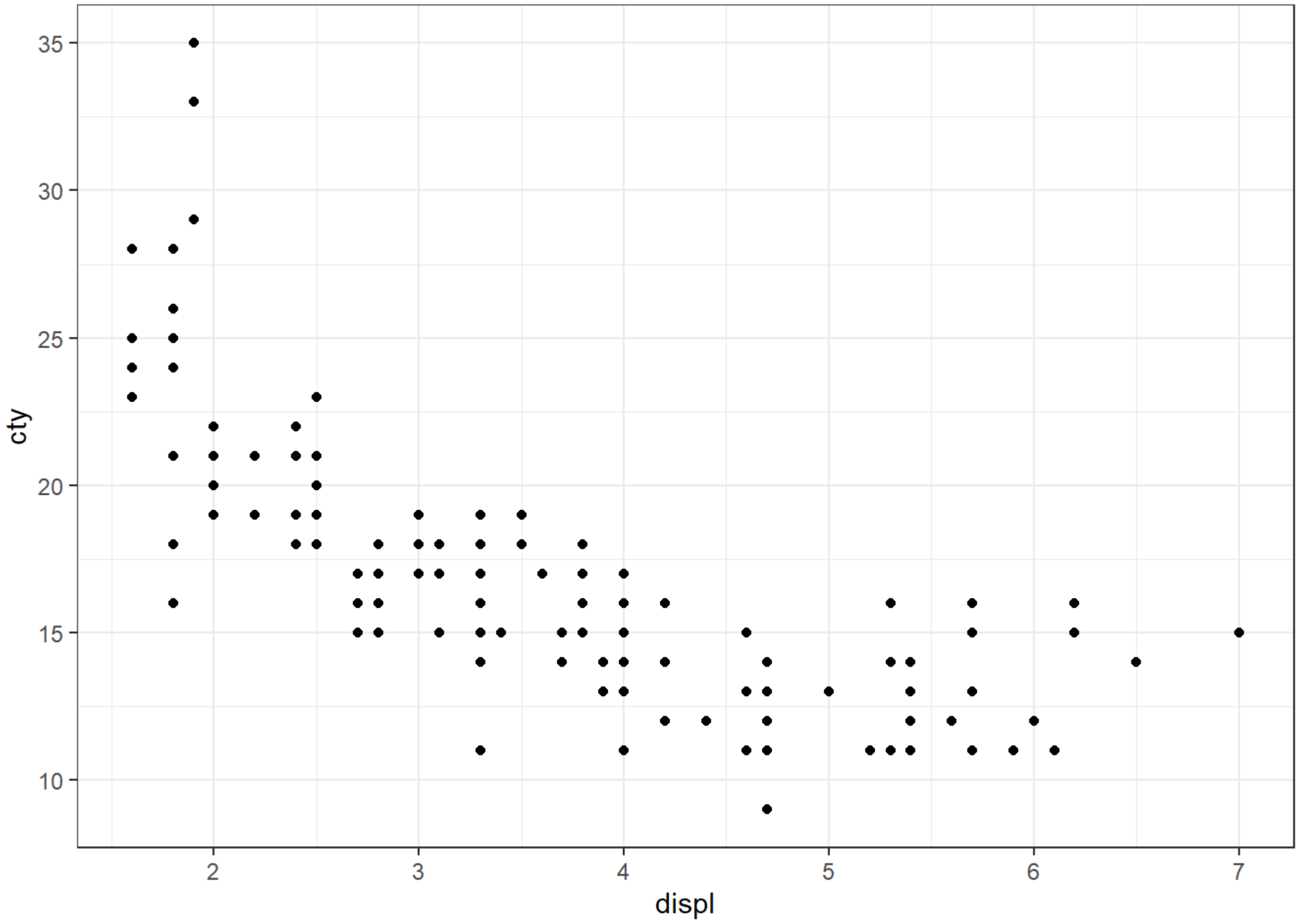
Were Tufte's "rules" met? Specifically his rule about the number of dimensions. What does shape = class do? Which is better, shape = class or color = class? What about combining shapes and colors? Is this a good idea? themes?

Ans - Tufte's rule is met if we avoid unnecessary encodings. `shape = class` assigns different symbols per class; `color = class` is usually clearer, especially with many groups. Shape works for few classes or grayscale. Combining both adds redundancy but can clutter. Minimal themes highlight data.

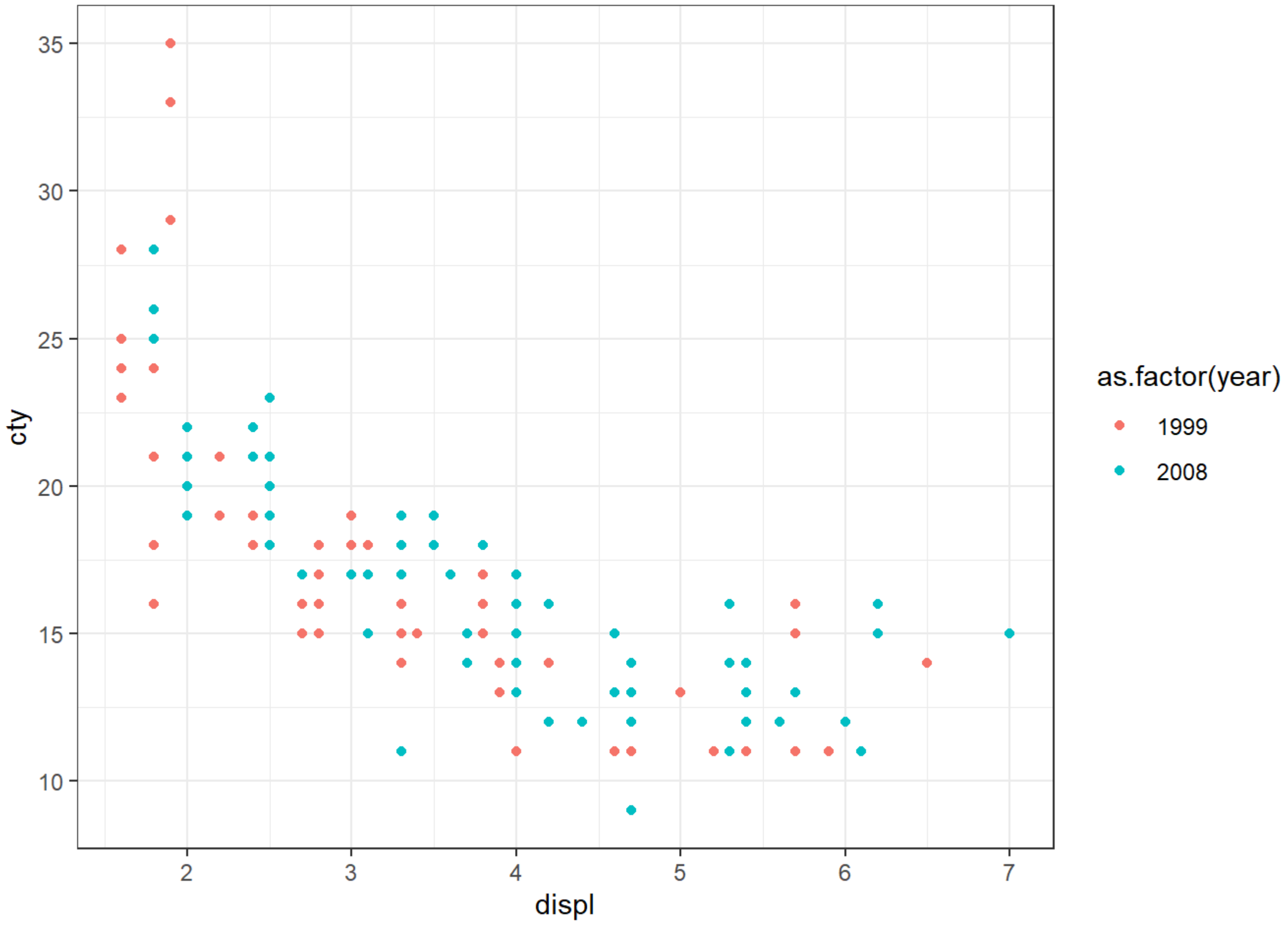
```
# Let's tweak things around!
# What does class %in% c(...) do?
# What does labs(....) do?
g <- ggplot(data = subset(mpg,
class %in% c("compact",
"midsize", "subcompact",
"suv")))
g + geom_point(aes(x=displ,
y=hwy, shape = class)) + labs(x="Displacement", y="Highway MPG")
```



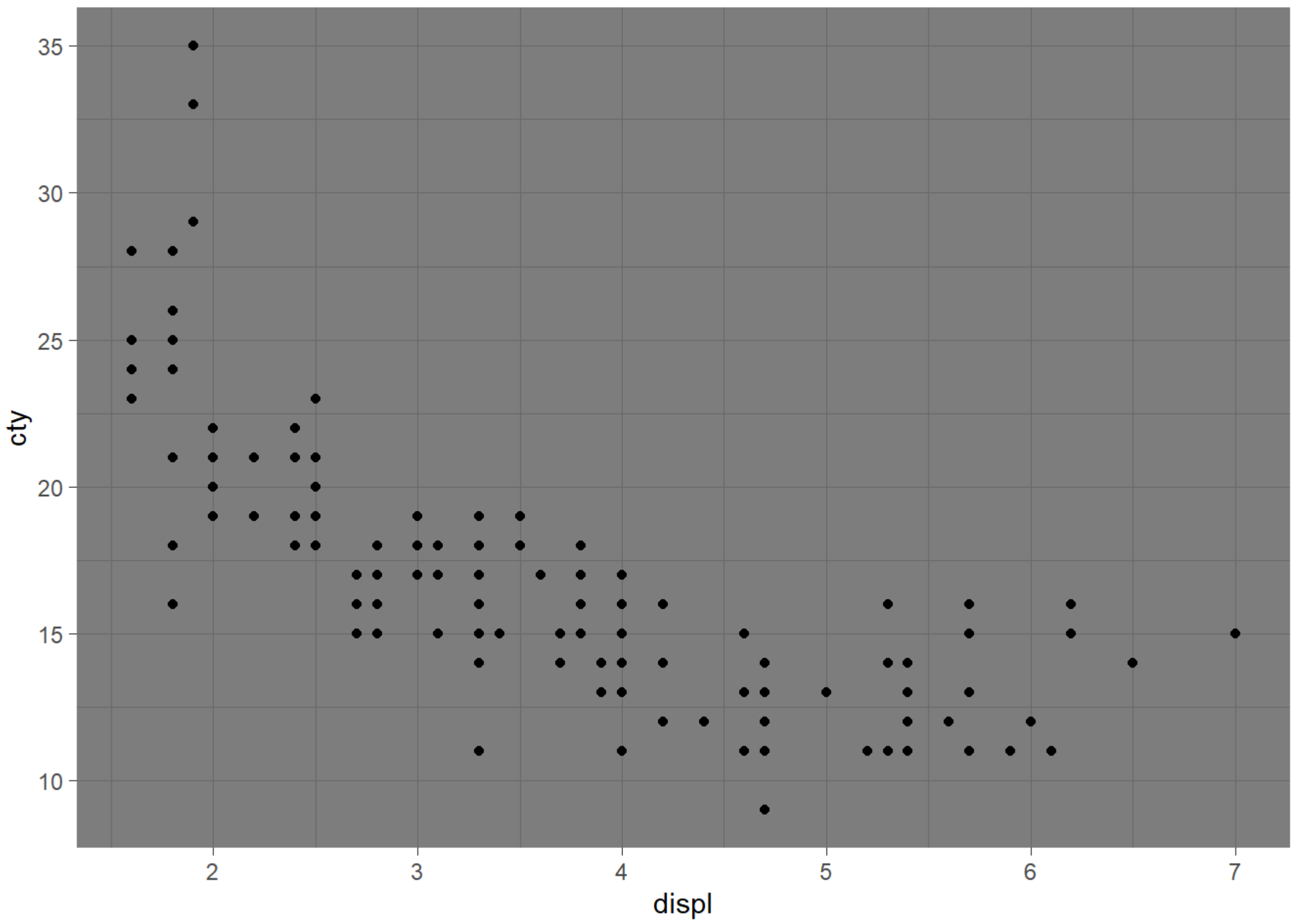

```
g <- ggplot(data = mpg,
mapping=aes(x=displ,
y=cty)) + theme_bw()
g+geom_point()
```



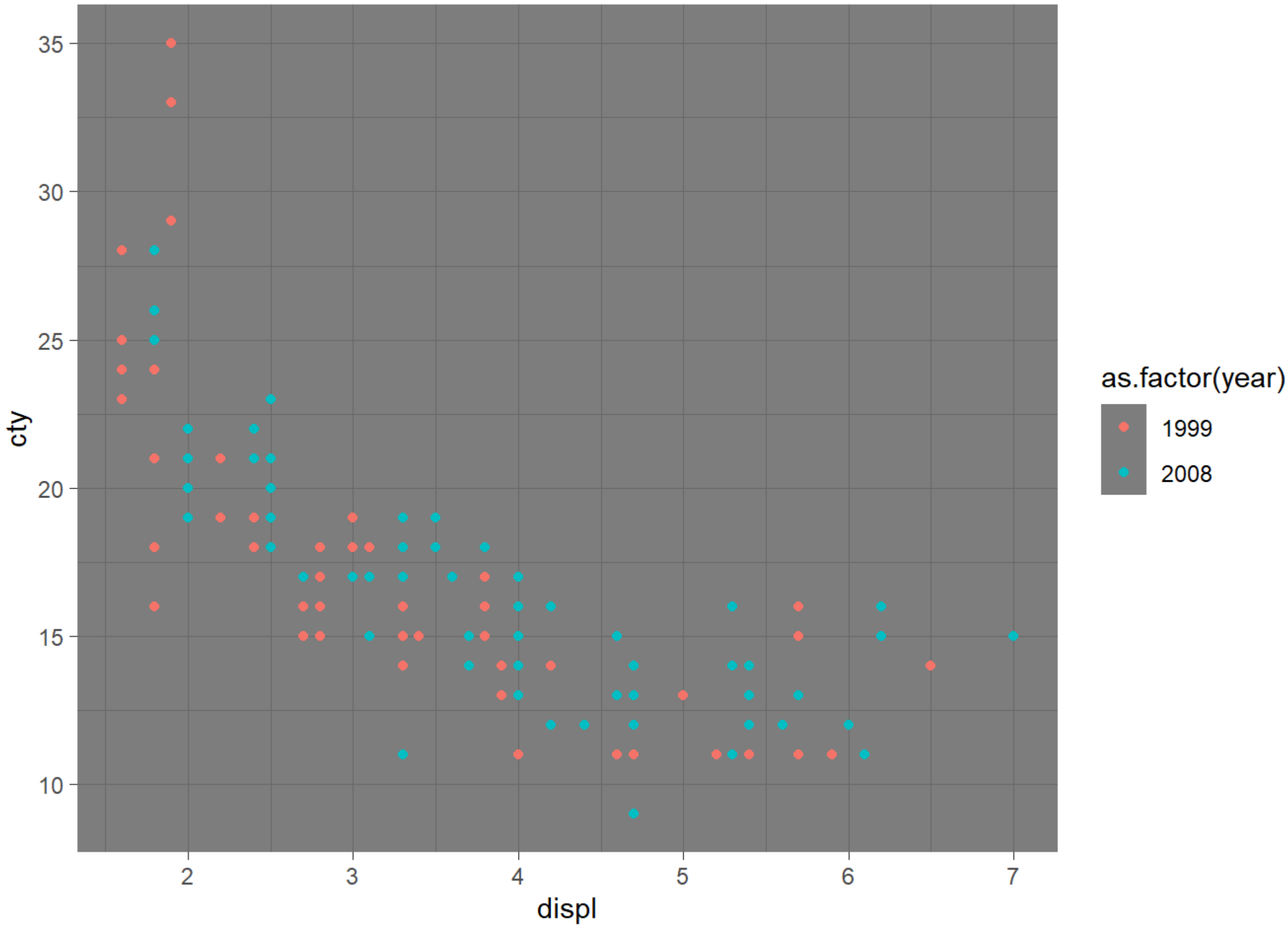

```
g + geom_point(aes(color=as.factor(year)))
```



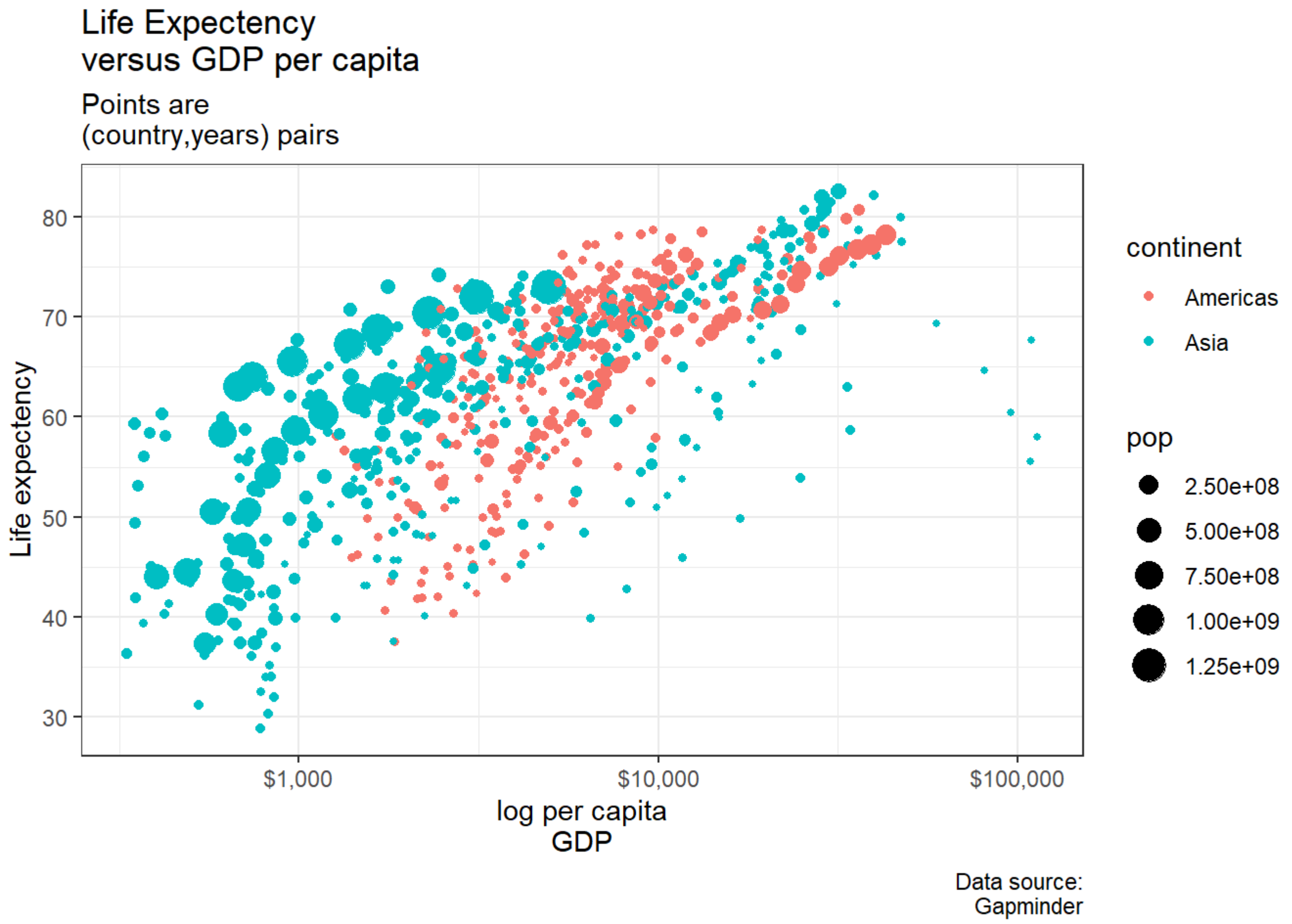
```
g <- ggplot(data = mpg,
mapping=aes(x=displ,
y=cty)) + theme_dark()
g+geom_point()
```



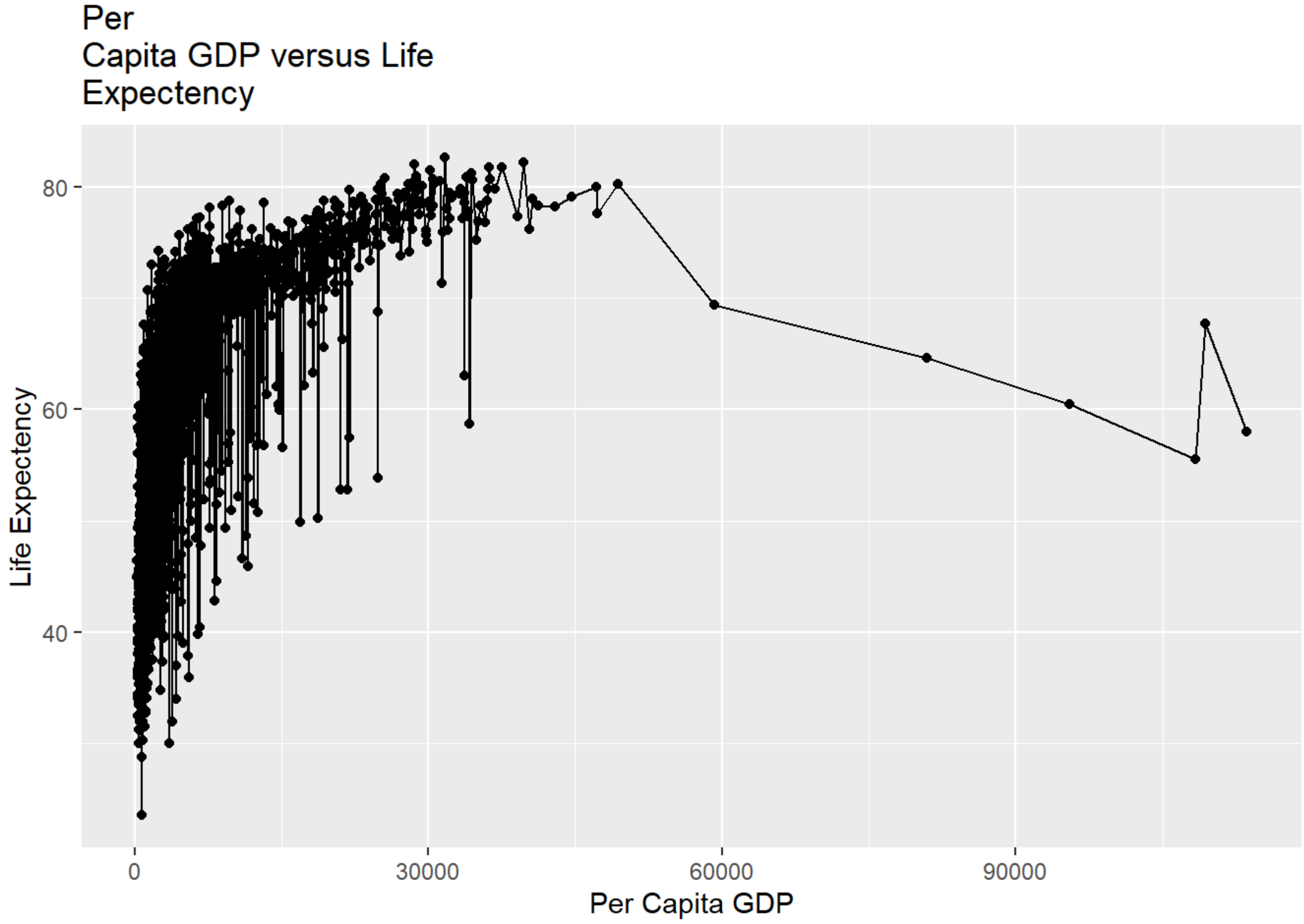

```
g + geom_point(aes(color=as.factor(year)))
```



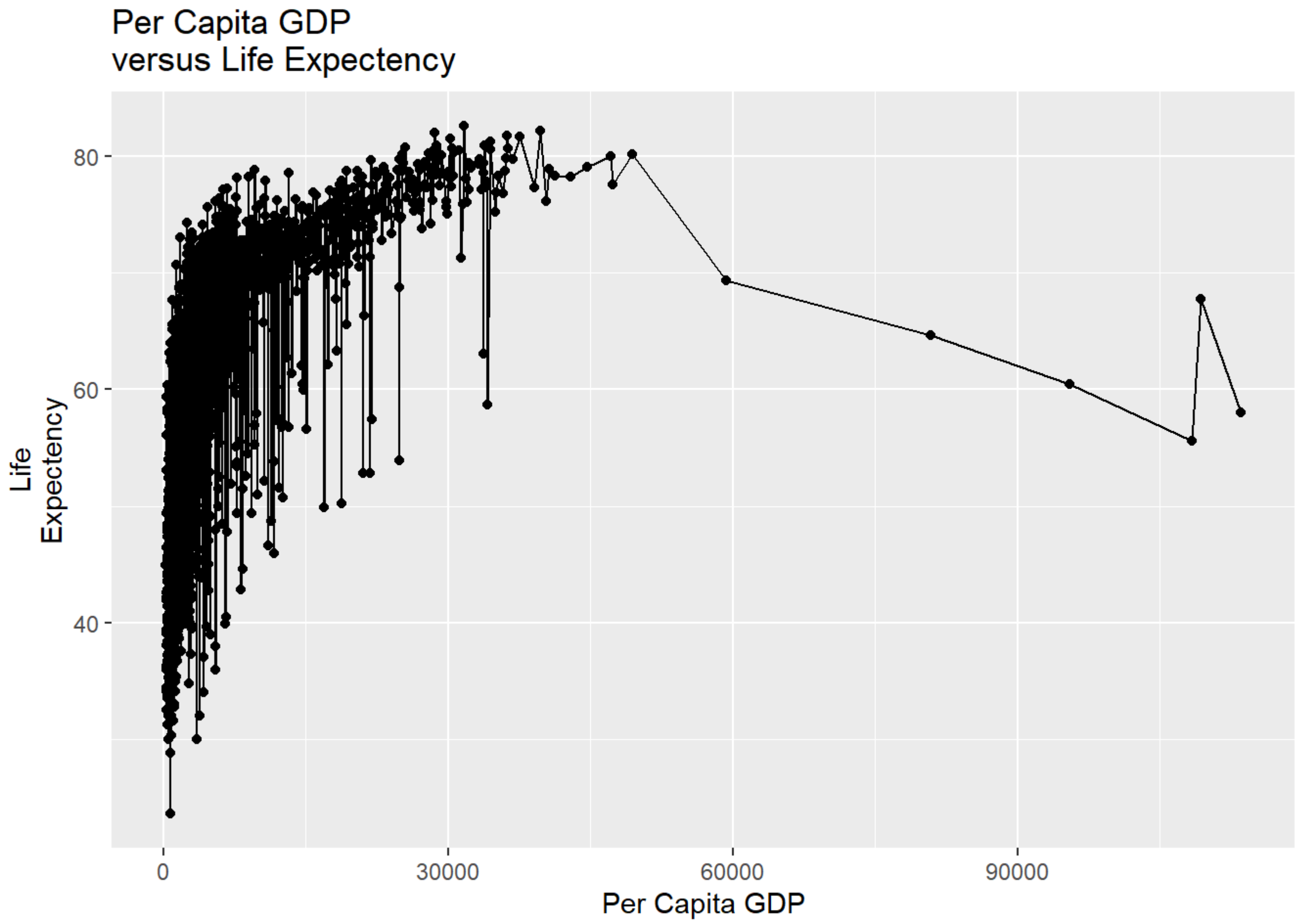

```
library(gapminder)

ggplot(data =
subset(gapminder,continent
%in% c("Asia", "Americas")),
mapping = aes(x = gdpPercap,
y = lifeExp)) +
theme_bw() +
geom_point(aes(size = pop, color
= continent)) +
coord_cartesian() +
scale_x_log10(labels =
scales::dollar) +
labs(x="log per capita
GDP",y="Life expectency",
title="Life Expectency
versus GDP per capita",
subtitle = "Points are
(country,years) pairs",
caption = "Data source:
Gapminder")
```



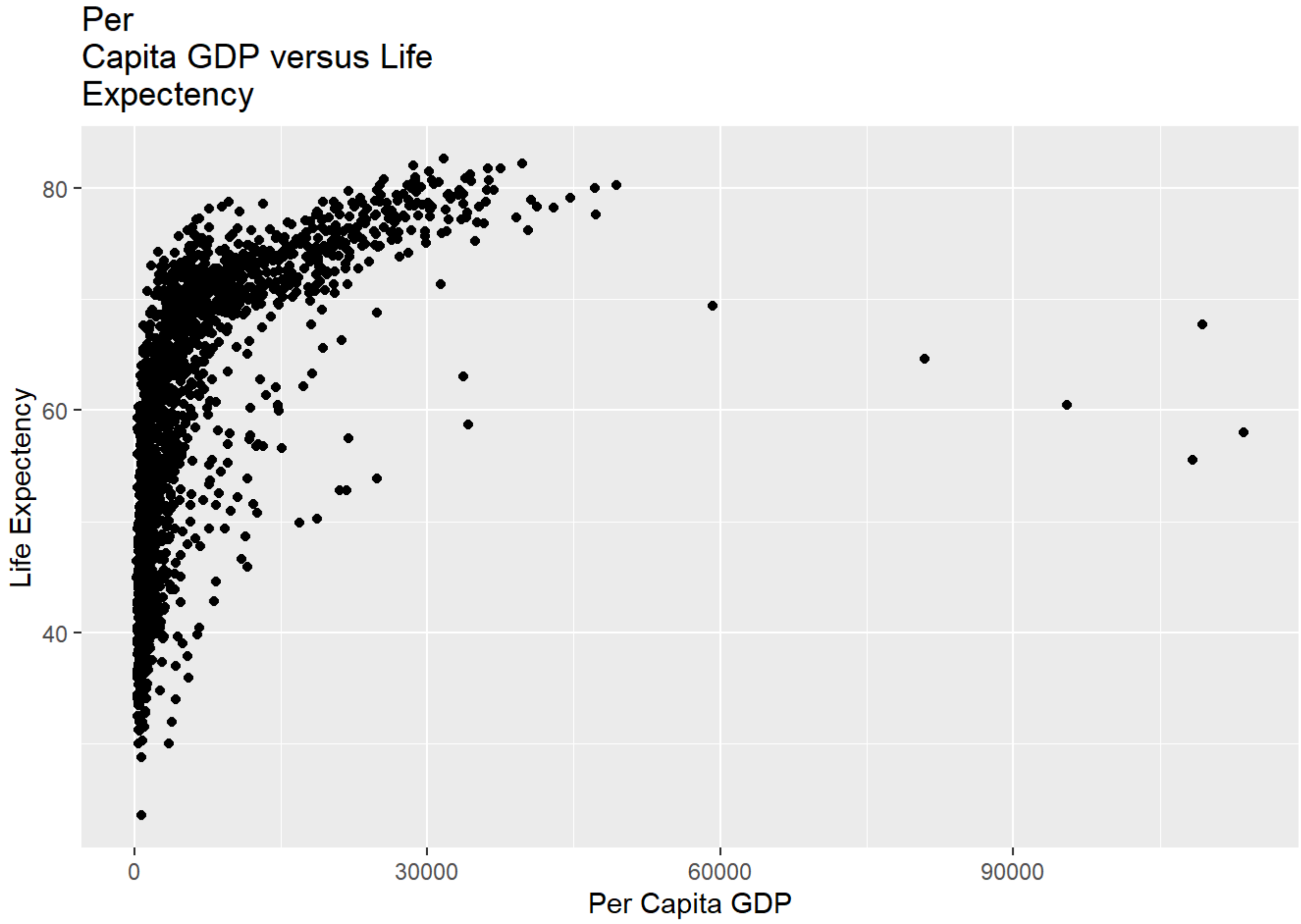

```
g <- ggplot(data=gapminder)
g+ geom_point(mapping = aes(x =
gdpPercap, y = lifeExp))+
geom_line(mapping = aes(x =
gdpPercap, y = lifeExp)) +
labs(x="Per Capita GDP",
y="Life Expectency", title="Per
Capita GDP versus Life
Expectency")
```




```
g <- ggplot(data=gapminder)
g+ geom_point(mapping = aes(x =
gdpPercap, y = lifeExp))+
geom_line(mapping = aes(x =
gdpPercap, y = lifeExp)) +
labs(x="Per Capita GDP", y="Life
Expectency", title="Per Capita GDP
versus Life Expectency")
```




```
g <- ggplot(data=gapminder)
g+ geom_point(mapping = aes(x =
gdpPercap, y = lifeExp)) +
labs(x="Per Capita GDP",
y="Life Expectency", title="Per
Capita GDP versus Life
Expectency")
```

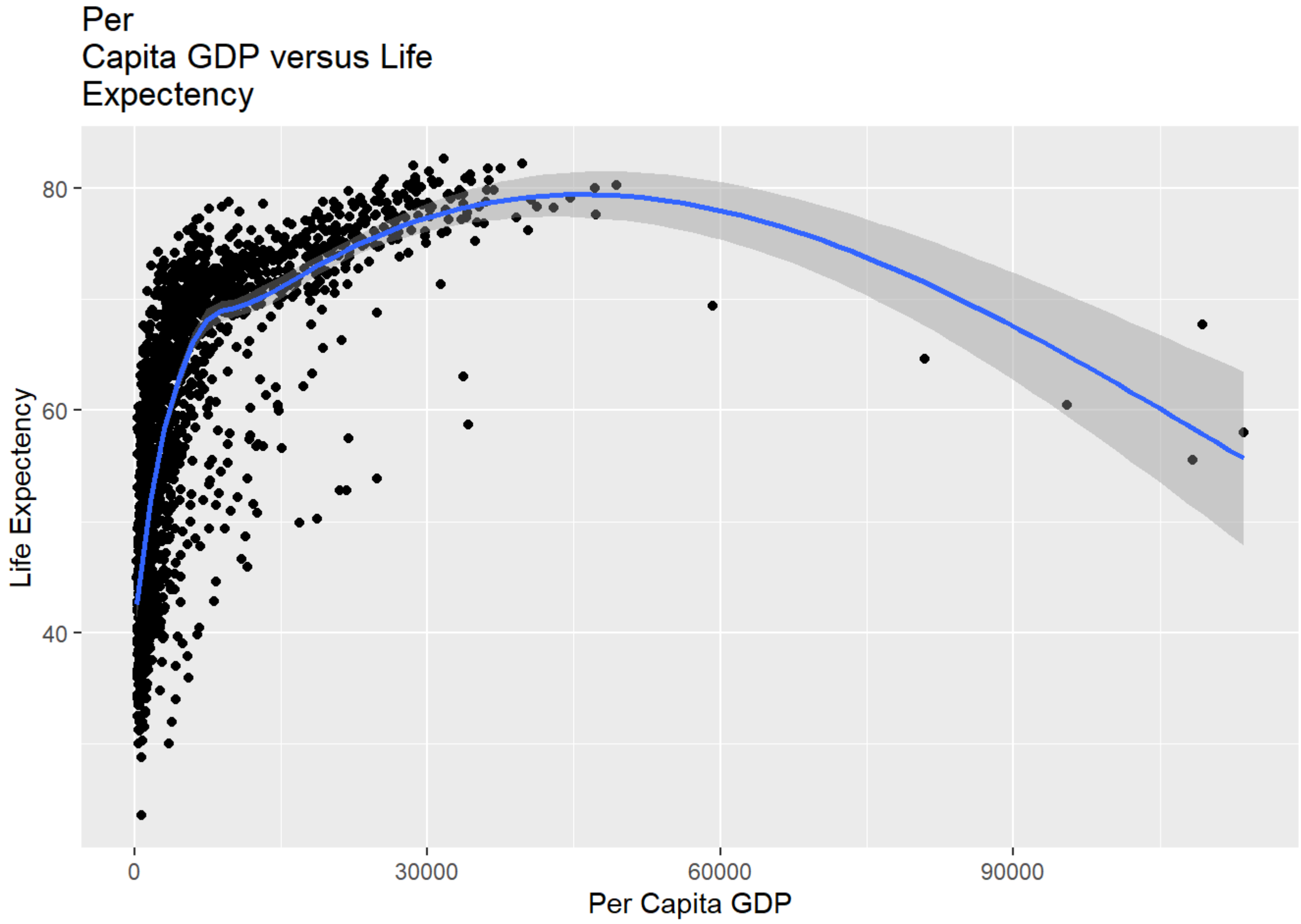


What are some issues with the graphic generated by the above code ?

The graphic may have overlapping points (overplotting), an x-axis that is skewed due to GDP values not being log-scaled, a misspelled y-axis label (“Life Expectency”), redundant aesthetic mappings, and the geom_smooth() curve may not appropriately represent the trend.

```
g <- ggplot(data=gapminder)
g+ geom_point(mapping = aes(x =
gdpPercap, y = lifeExp)) +
geom_smooth(mapping = aes(x =
gdpPercap, y = lifeExp)) +
labs(x="Per Capita GDP",
y="Life Expectency", title="Per
Capita GDP versus Life
Expectency")
```

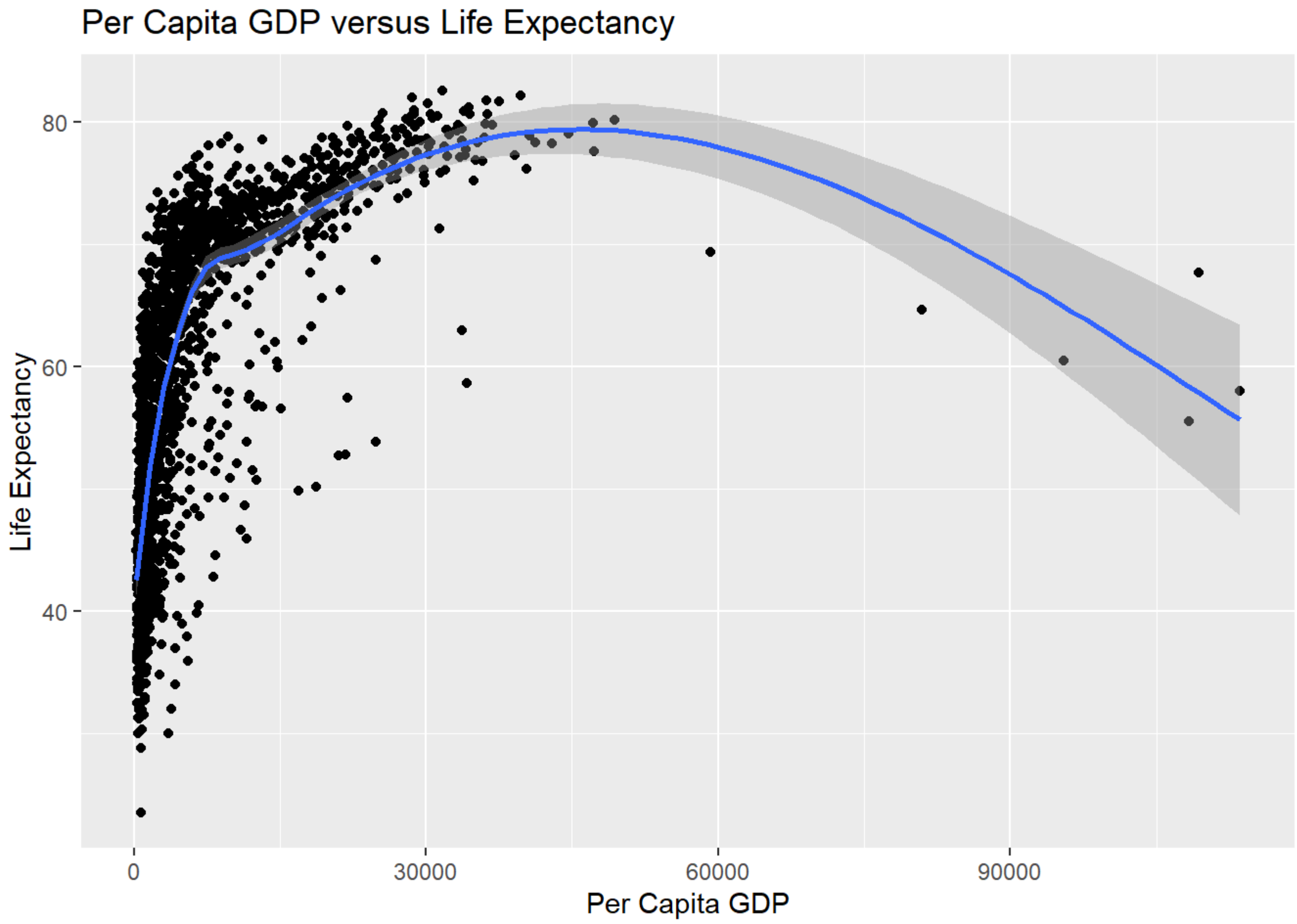
```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```




```
## `geom_smooth()` using method = 'gam' and formula 'y ~ s(x,bs = "cs")'
```

```
g <- ggplot(data = gapminder)
g +
  geom_point(mapping = aes(x = gdpPercap, y = lifeExp)) +
  geom_smooth(mapping = aes(x = gdpPercap, y = lifeExp)) +
  labs(
    x = "Per Capita GDP",
    y = "Life Expectancy",
    title = "Per Capita GDP versus Life Expectancy"
  )
```

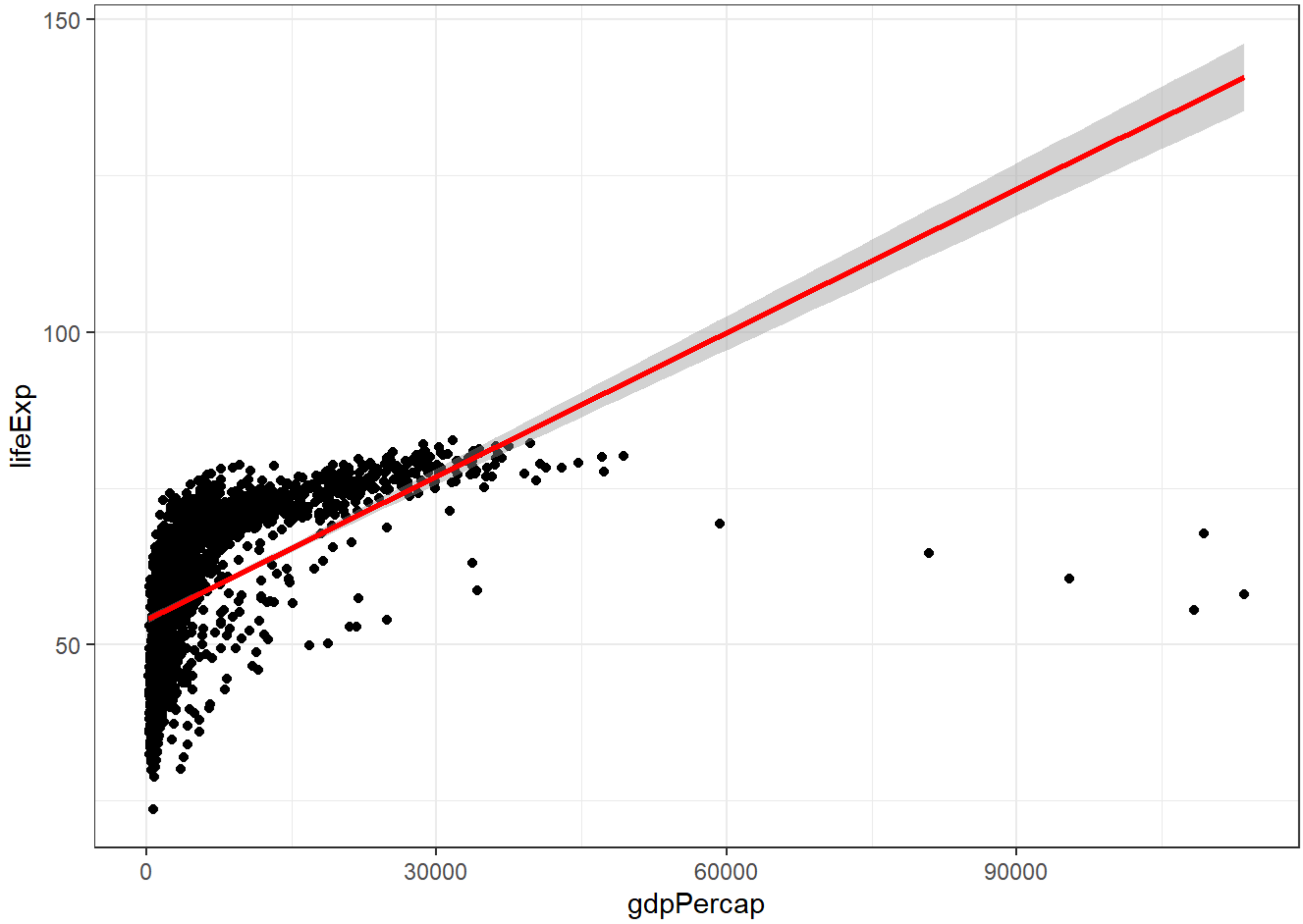
```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```




```
## `geom_smooth()` using method = 'gam' and formula 'y ~ s(x,bs = "cs")'
```

```
ggplot(data=gapminder,
aes(x = gdpPercap,
y = lifeExp))+
geom_point( ) +
geom_smooth(color=
"red",method="lm")+
theme_bw( )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula 'y ~ x'
```

What is the frequentlist interpretation of probability? Explain the following line code and how it relates to this

Ans - The frequentist view sees probability as the long-run relative frequency of an event when the same experiment is repeated under identical conditions. In this sense, probability is not about personal belief or opinion, but about the stable proportion of times an outcome occurs across many trials.

```
y <- rbinom(1000000, 1, .5)
sum(y)
```

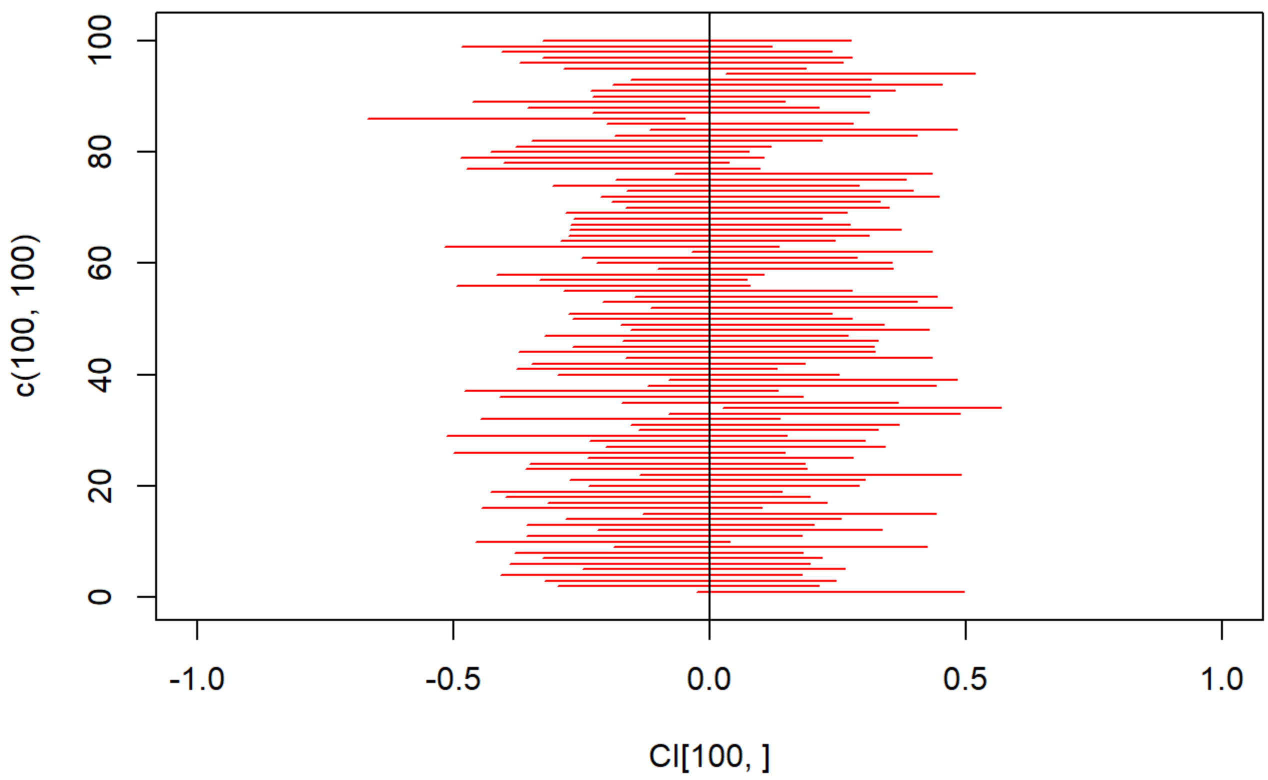
```
## [1] 499983
```

How does the following picture convey the role of a conifdence interval ? Hint you answer should explain how some CIs do not contain the parameter and some do.

Ans - The plot conveys the role of a **confidence interval (CI)** by illustrating a simulation. Each red horizontal line is a **CI** calculated from a different sample.

The vertical line at **0.0** represents the true **population parameter** (often denoted as θ or μ). The CIs that **cross** the vertical line (the majority of them) **contain** the true parameter. The CIs that **do not cross** the vertical line (a small number of them, e.g., the lowest one) **fail to contain** the true parameter. This visual demonstrates that a CI provides a range of plausible values for a parameter, but in repeated sampling, a specific percentage (e.g., 5%) of the CIs is expected to miss the true value.

```
set.seed(4)
CI<-matrix(0,100,2)
for(i in 1:100){
x<-rnorm(50,0,1)
CI[i,]<-mean(x)+c(qt(0.025,
49)*sd(x)/sqrt(50), qt(0.975,
49)*sd(x)/sqrt(50))
}
plot(CI[100,],c(100,100),type="l",col=
"red",
ylim=c(0,101), xlim=c(-1,1))
for(i in 1:99){
lines(CI[i,],c(i,i),type="l",col="red")
}
abline(v=0, col="black")
```




```
sum(CI[,1]<0 & CI[,2]>0)
```

```
## [1] 97
```

```
library(gapminder)
library(plyr)
```

```
## ------------------------------------------------------------------------------
```

```
## You have loaded plyr after dplyr - this is likely to cause problems.
## If you need functions from both plyr and dplyr, please load plyr first, then dplyr:
## library(plyr); library(dplyr)
```

```
## ------------------------------------------------------------------------------
```

```
##
## Attaching package: 'plyr'
```

```
## The following objects are masked from 'package:dplyr':
##
##     arrange, count, desc, failwith, id, mutate, rename, summarise,
##     summarize
```

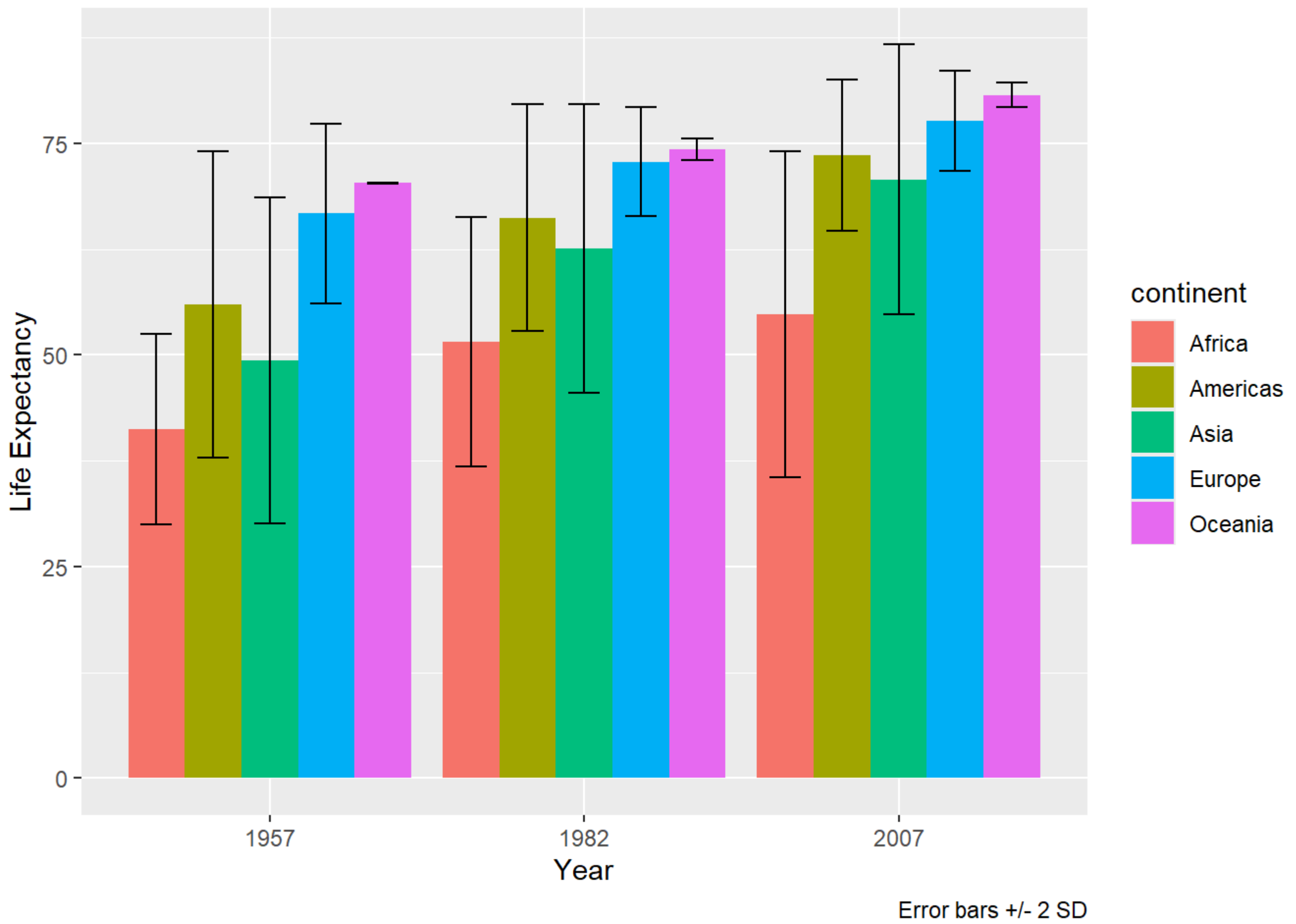
```
## The following object is masked from 'package:purrr':
##
##     compact
```

```
gapminder2 <- ddply(gapminder, ~ year + continent, summarize,
MeanLE = mean(lifeExp), UCI =
mean(lifeExp)+2*sd(lifeExp), LCI = mean(lifeExp)-
2*sd(lifeExp))

gapminder2
```

```
##    year continent   MeanLE      UCI      LCI
## 1  1952    Africa 39.13550 49.43866 28.83234
## 2  1952  Americas 53.27984 71.93200 34.62768
## 3  1952      Asia 46.31439 64.89790 27.73089
## 4  1952    Europe 64.40850 77.13068 51.68632
## 5  1952   Oceania 69.25500 69.63684 68.87316
## 6  1957    Africa 41.26635 52.50659 30.02610
## 7  1957  Americas 55.96028 74.02666 37.89390
## 8  1957      Asia 49.31854 68.58940 30.04769
## 9  1957    Europe 66.70307 77.29468 56.11146
## 10 1957   Oceania 70.29500 70.39399 70.19601
## 11 1962    Africa 43.31944 55.07017 31.56871
## 12 1962  Americas 58.39876 75.40585 41.39167
## 13 1962      Asia 51.56322 71.20449 31.92196
## 14 1962    Europe 68.53923 77.14423 59.93423
## 15 1962   Oceania 71.08500 71.52341 70.64659
## 16 1967    Africa 45.33454 57.49988 33.16919
## 17 1967  Americas 60.41092 76.22926 44.59258
## 18 1967      Asia 54.66364 73.96557 35.36171
## 19 1967    Europe 69.73760 77.33706 62.13814
## 20 1967   Oceania 71.31000 71.90397 70.71603
## 21 1972    Africa 47.45094 60.28346 34.61843
## 22 1972  Americas 62.39492 77.04095 47.74889
## 23 1972      Asia 57.31927 76.76467 37.87387
## 24 1972    Europe 70.77503 77.25619 64.29388
## 25 1972   Oceania 71.91000 71.96657 71.85343
## 26 1977    Africa 49.58042 63.19682 35.96403
## 27 1977  Americas 64.39156 78.53055 50.25257
## 28 1977      Asia 59.61056 79.65495 39.56616
## 29 1977    Europe 71.93777 78.17983 65.69571
## 30 1977   Oceania 72.85500 74.65105 71.05895
## 31 1982    Africa 51.59287 66.34475 36.84099
## 32 1982  Americas 66.22884 79.67051 52.78717
## 33 1982      Asia 62.61794 79.68838 45.54750
## 34 1982    Europe 72.80640 79.24292 66.36988
## 35 1982   Oceania 74.29000 75.56279 73.01721
## 36 1987    Africa 53.34479 69.07297 37.61661
## 37 1987  Americas 68.09072 79.69458 56.48686
## 38 1987      Asia 64.85118 81.25877 48.44360
## 39 1987    Europe 73.64217 79.98153 67.30281
## 40 1987   Oceania 75.32000 78.14843 72.49157
## 41 1992    Africa 53.62958 72.55172 34.70743
## 42 1992  Americas 69.56836 79.90257 59.23415
## 43 1992      Asia 66.53721 82.68831 50.38611
## 44 1992    Europe 74.44010 80.85966 68.02054
## 45 1992   Oceania 76.94500 78.68448 75.20552
## 46 1997    Africa 53.59827 71.80504 35.39150
## 47 1997  Americas 71.15048 80.92565 61.37531
## 48 1997      Asia 68.02052 84.20286 51.83817
## 49 1997    Europe 75.50517 81.71452 69.29581
## 50 1997   Oceania 78.19000 80.00019 76.37981
## 51 2002    Africa 53.32523 72.49822 34.15224
## 52 2002  Americas 72.42204 82.02145 62.82263
## 53 2002      Asia 69.23388 85.98307 52.48469
## 54 2002    Europe 76.70060 82.54496 70.85624
```

```
## 55 2002   Oceania 79.74000 81.52191 77.95809
## 56 2007    Africa 54.80604 74.06760 35.54448
## 57 2007  Americas 73.60812 82.49002 64.72622
## 58 2007      Asia 70.72848 86.65593 54.80104
## 59 2007    Europe 77.64860 83.60823 71.68897
## 60 2007   Oceania 80.71950 82.17755 79.26145
```

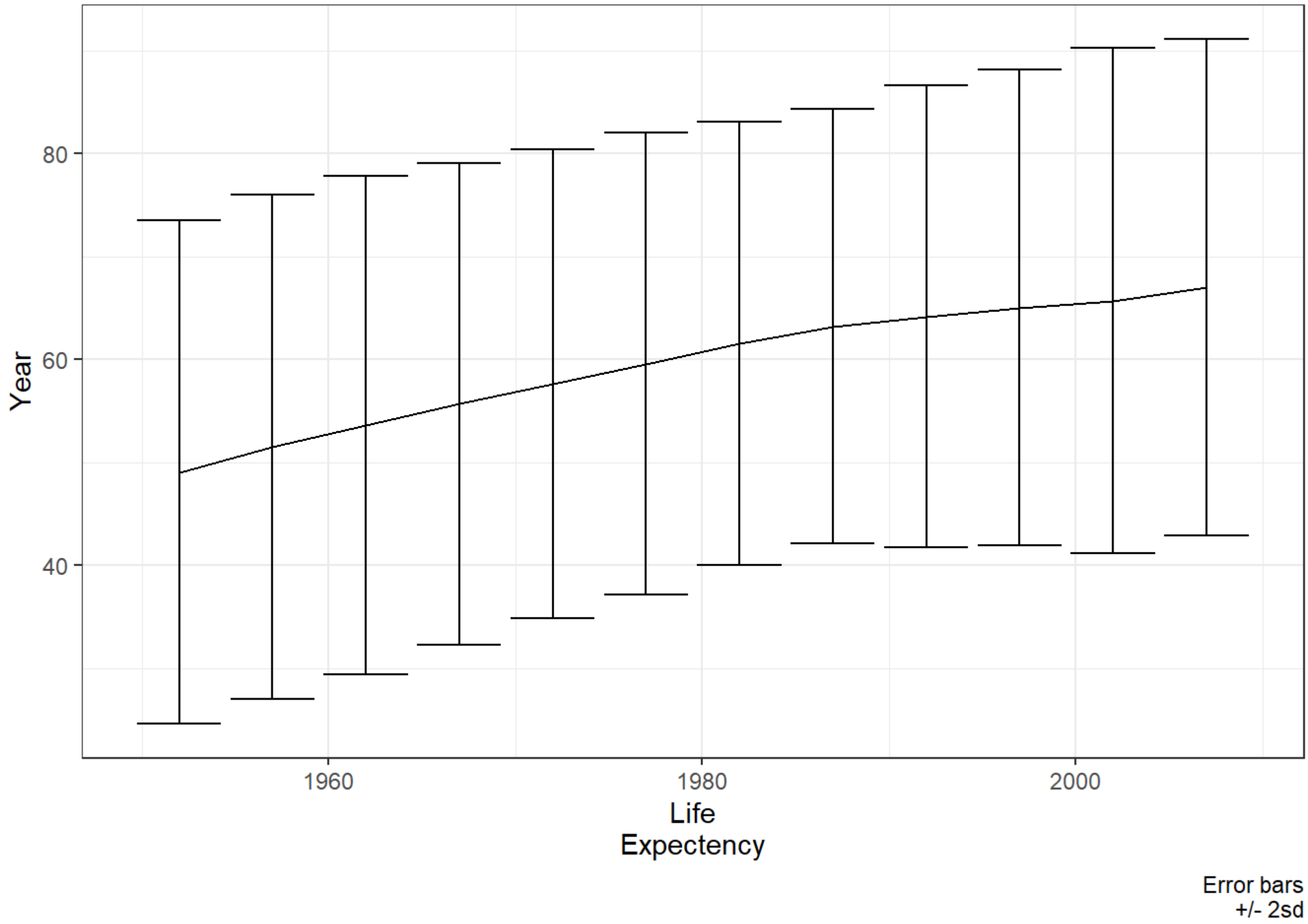
```
p <- ggplot(
  data = subset(gapminder2, year %in% c(1957, 1982, 2007)),
  aes(x = as.factor(year), y = MeanLE)
)

p +
  geom_bar(aes(fill = continent),
           stat = "identity",
           position = position_dodge()) +
  geom_errorbar(aes(ymin = LCI, ymax = UCI, group = continent),
                width = 0.5,
                position = position_dodge(0.9)) +
  labs(
    x = "Year",
    y = "Life Expectancy",
    caption = "Error bars +/- 2 SD"
  )
```



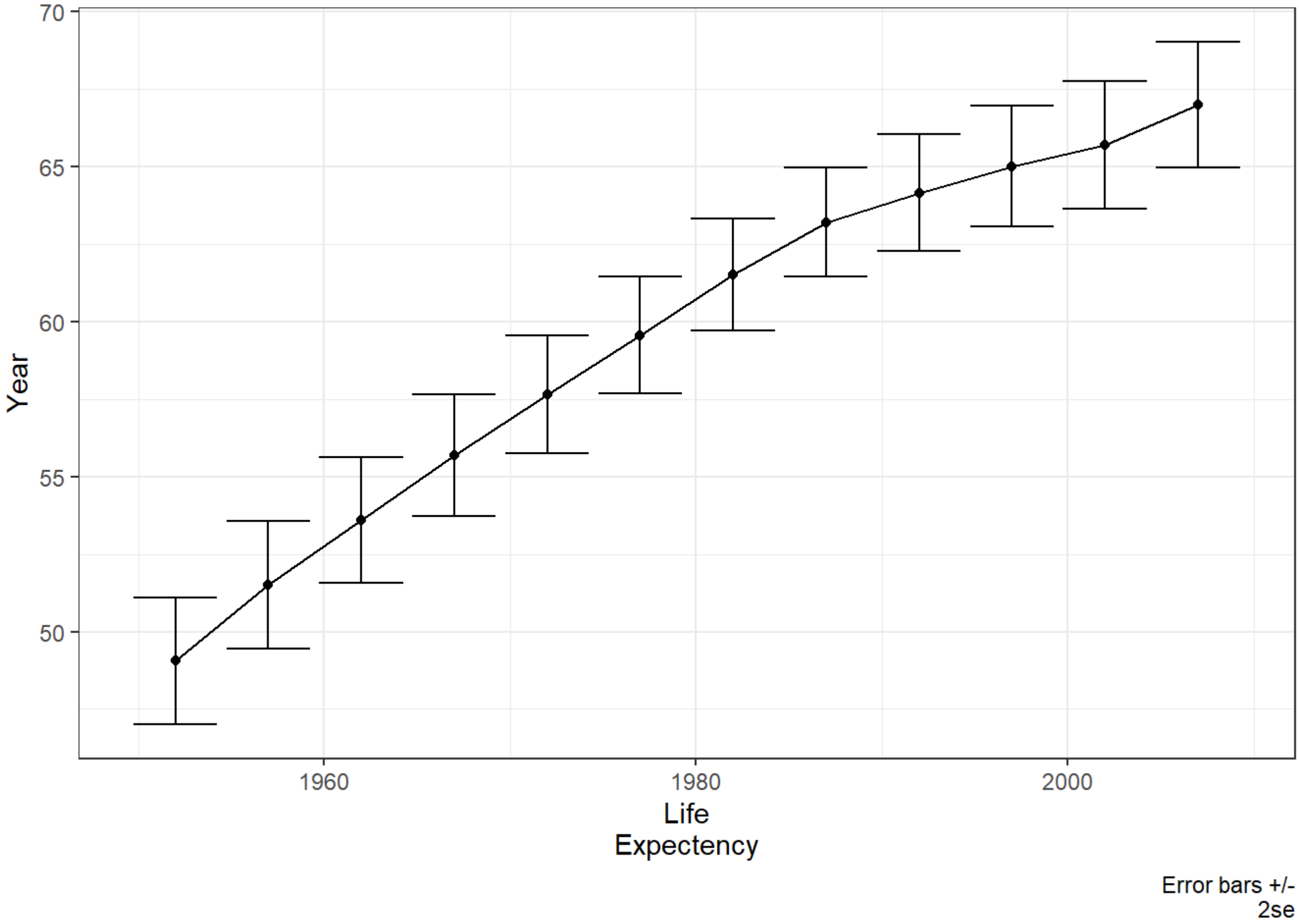
```
gapminder3 <- ddply(gapminder, ~ year, summarize,
MeanLE = mean(lifeExp),
UCIsd = mean(lifeExp)+2*sd(lifeExp),
UCIse = mean(lifeExp)+2*sd(lifeExp)/sqrt(length(lifeExp)),
LCIsd = mean(lifeExp)-2*sd(lifeExp),
LCIse = mean(lifeExp)-2*sd(lifeExp)/sqrt(length(lifeExp)))

head(gapminder3)
```

```
##   year   MeanLE    UCIsd    UCIse    LCIsd    LCIse
## 1 1952 49.05762 73.50953 51.10958 24.60571 47.00566
## 2 1957 51.50740 75.96997 53.56025 27.04483 49.45455
## 3 1962 53.60925 77.80374 55.63961 29.41476 51.57889
## 4 1967 55.67829 79.11601 57.64514 32.24057 53.71144
## 5 1972 57.64739 80.41129 59.55769 34.88348 55.73708
## 6 1977 59.57016 82.02462 61.45449 37.11570 57.68582
```

```
p<- ggplot(data=
gapminder3,
aes(x=year,y=MeanLE))+
theme_bw()
p + geom_line( ) +
geom_errorbar(aes(
ymin=LCIsd,
ymax=UCIsd)) +
labs(y="Year", x="Life
Expectency",
caption="Error bars
+/- 2sd")
```




```
p<- ggplot(data=gapminder3,
aes(x=year, y=MeanLE)) +
theme_bw()
p + geom_line( ) +
geom_point( ) +
geom_errorbar(aes(ymin=
LCIse, ymax=UCIse))+
labs(y="Year", x="Life
Expectency",
caption="Error bars +/-
2se")
```


70
65
60
55
50
Year
1960
1980
2000
Life
Expectency
Error bars +/-
2se